

Michael C. Anoruo

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EDUCATION

Carnegie Mellon University

Doctor of Philosophy in Robotics
Advisor: Dr. Wennie Tabib

Pittsburgh, PA
Anticipated May 2029

University of Maryland, Baltimore County

Bachelor of Science in Computer Science — AI/ML Track
GPA: 3.93/4.0, Summa Cum Laude

Baltimore, MD
Dec 2023

GRADUATE COURSES

Learning for 3D Vision	Deep Reinforcement Learning
Optimal Control & Reinforcement Learning	Robot Learning
Mobile Robots	Computer Vision
Math Fundamentals for Robotics	Data Science

SKILLS

Languages: Python, Lua, C++, C
Robotics & Hardware: ROS, ROS 2, Vicon, Crazyflie, KiCad
Deep Learning & RL: PyTorch, TensorFlow, Gymnasium
Computer Vision: OpenCV, Open3D, PyTorch3D
Build Tools & Automation: CMake, Colcon

PHD RESEARCH

Carnegie Mellon University — Resilient Intelligent Systems Lab (RISLab)
Graduate Researcher

Pittsburgh, PA
Aug 2024 – Present

- Developed a differentiable, GPU-accelerated quadrotor simulator in PyTorch capable of simulating 100k+ drones in parallel on a single NVIDIA RTX 5090.
- Built a Gymnasium environment to train RL quadrotor control policies for dynamic target interception, achieving over 95% success against targets moving up to 10 m/s.
- Integrated third-party monocular cameras onto Crazyflie drones by wiring and soldering the camera modules, enabling live wireless video streaming to a desktop for offboard processing.
- Adapted the lab's Crazyflie flight stack to use Vicon motion-capture data, enabling multi-drone swarm experiments.
- Deployed a PostgreSQL-backed MLflow server behind Apache, centralizing experiment tracking and providing on-demand access to training results for a five-person research team.

RESEARCH EXPERIENCE

Johns Hopkins University APL — Asymmetric Operations Sector
Software Engineer Intern

Baltimore, MD
May 2024 – Aug 2024

- Enhanced a GUI module and contributed to an integration layer in Java/JavaFX for a secure communication system, interfacing with encryption APIs to transmit data as part of a large engineering team.
- Led a team of 7 interns on a drone swarming project for search-and-rescue, researching and testing A*-based search algorithms in PyBullet simulation.
- Trained a variational autoencoder on network packet data to support anomaly-based network intrusion detection.
- Built an internal logging service using PostgreSQL and Flask for network monitoring, saving the company hundreds of thousands of dollars in Splunk licensing fees.

Massachusetts Institute of Technology — Space Enabled Media Lab
Zero Robotics Research Intern

Cambridge, MA
Jun 2022 – Aug 2022

- Helped design technical challenges for Zero Robotics, an international programming competition in which students write code that is executed on robots aboard the International Space Station.
- Tuned PID gains in the MIT Astrobee simulator to better align simulated trajectory timing with the physical platform in space, benchmarking performance across multiple reference trajectories.
- Collaborated with a team of six, alongside NASA astronauts, to host the Zero Robotics 2022 final competition.

- Helped restore the lab's Clearpath Heron unmanned surface vehicle (USV), testing the bathymetric sensor and validating autonomy by collecting and replaying ROS bag files from field experiments.
- Designed a waypoint-generation algorithm that produced lawnmower survey paths within user-defined convex regions for the lab's USV to map the Schuylkill riverbed.
- Built and validated three RS-232-to-USB interface boards, handling KiCad design, component sourcing, fabrication, soldering, and testing to enable data transfer from the Heron USV's onboard computer to a laptop.
- Built a Python simulator to test path-planning strategies for coordinated teams of miniature USVs pushing floating objects to specified goal locations.
- Repaired miniature USVs by wiring motors and flashing Arduino controllers, then integrated OptiTrack localization for physical collaborative object-pushing experiments.

PROFESSIONAL EXPERIENCE

Apple Inc. — Apple Service Engineering
Data Engineer Intern

Seattle, WA
May 2023 – Aug 2023

- Designed and built an end-to-end computer-vision system that segmented video, analyzed per-segment viewership, and generated sponsor reports, reducing analysis time from over 3 hours to under 10 minutes.
- Presented the system architecture, performance results, and business impact to senior engineering leadership within Apple Service Engineering.

PUBLICATIONS

Dang, Q., Kucukosmanoglu, M., **Anoruo, M.**, Kargosha, G., Conklin, S., & Brooks, J. "Automatic Detection of Cognitive Events Using Machine Learning and Understanding Models' Interpretations of Human Cognition." *Scientific Reports* 15, no. 1 (2025): 30506.

Piplai, A., **Anoruo, M.**, Fasaye, K., Joshi, A., Finin, T., & Ridley, A. "Knowledge Guided Two-Player Reinforcement Learning for Cyber Attacks and Defenses." *2022 21st IEEE International Conference on Machine Learning and Applications (ICMLA)* (2022): 1342–1349.

CONFERENCES / PRESENTATIONS

NVIDIA Day Robotics Research Talk, Carnegie Mellon University

Pittsburgh, PA, May 2026

Apple Service Engineering Intern Spotlight, Apple Inc.

Seattle, WA, Aug 2023

MSRP Summer Research Symposium, Massachusetts Institute of Technology

Cambridge, MA, Aug 2022

On-Ramp Mission Research Symposium, National Security Agency

Baltimore, MD, Mar 2022

SUNFEST Symposium, University of Pennsylvania

Philadelphia, PA, Aug 2021

TEACHING ASSISTANT EXPERIENCE

Carnegie Mellon University

Pittsburgh, PA

Teaching Assistant — 16-761 Mobile Robots

Spring 2026

Teaching Assistant — 16-362 Mobile Robot Algorithms Laboratory

Fall 2025

LEADERSHIP

Project Lead, AI Scholars Program, CMU

Summer 2026

Site Lead, Creative Coders, UMBC

Spring 2023

Peer Advisor, Meyerhoff Scholars Program, UMBC

Fall 2022 – Spring 2023

Executive Board, National Society of Black Engineers, UMBC

Fall 2021 – Spring 2023

AWARDS & HONORS

GEM Fellow

Fall 2024

Google CSRMP Scholar

Spring 2022

Meyerhoff Scholar

Fall 2020

National Football Foundation Scholar Athlete

Spring 2020